UNIT 7 · ARS/NET PATTERN

MCQ Bank — 140 Questions with Answers

Energy sources; solar radiation, spectrum and measurement (Q1–Q35)

Q1. Which is a CONVENTIONAL (non-renewable) energy source?

(a) Solar (b) Wind
(c) Petroleum/diesel (d) Biogas

Q2. Which is a RENEWABLE energy source?

(a) Coal (b) Natural gas
(c) Nuclear (d) Solar

Q3. Wind, biomass and hydro energy are ultimately forms of:

(a) Nuclear energy (b) Indirect solar energy
(c) Tidal energy (d) Geothermal energy

Q4. The largest INDIRECT (embodied) energy input in crop production is usually:

(a) Seed (b) Nitrogen fertiliser
(c) Machinery (d) Pesticide

Q5. The solar constant is the irradiance on a normal surface at mean Earth–Sun distance:

(a) Inside the atmosphere (b) Just outside the atmosphere
(c) At the sun's surface (d) At the equator only

Q6. The accepted value of the solar constant (WMO) is about:

(a) 137 W/m² (b) 1367 W/m²
(c) 5800 W/m² (d) 100 W/m²

Q7. In cal/cm²·min, the solar constant is approximately:

(a) 0.2 (b) 1.94
(c) 10 (d) 50

Q8. The solar constant varies over the year by about:

(a) 0% (b) ±3.3%
(c) ±25% (d) ±50%

Q9. The sun radiates approximately as a black body at:

(a) 300 K (b) 1000 K
(c) 5800 K (d) 15,000 K

Q10. By Wien's law, the peak of the solar spectrum lies near:

(a) 0.10 µm (b) 0.48 µm
(c) 1.5 µm (d) 10 µm

Q11. The visible band carries roughly what fraction of solar energy?

(a) 7% (b) 25%
(c) 47% (d) 90%

Q12. The ultraviolet portion of the solar spectrum is about:

(a) 7% (b) 30%
(c) 50% (d) 70%

Q13. Photosynthetically active radiation (PAR) spans wavelengths:

(a) 0.1–0.2 μm (b) 0.4–0.7 μm

(c) 1–3 μm (d) 5–10 μm

Q14. Radiation coming straight from the sun's disc is called:

(a) Diffuse (b) Beam (direct)

(c) Reflected (d) Global

Q15. Global radiation on a horizontal surface equals:

(a) Beam only (b) Diffuse only

(c) Beam + diffuse (d) Beam - diffuse

Q16. Air mass is defined as:

(a) cos θ_z (b) 1/cos θ_z

(c) sin θ_z (d) tan θ_z

Q17. AM0 refers to:

(a) Sea level overhead (b) Extraterrestrial (outside atmosphere)

(c) 45° sun (d) Cloudy sky

Q18. The standard reference spectrum for rating PV modules is:

(a) AM0 (b) AM1

(c) AM1.5 (d) AM3

Q19. Atmospheric absorption of solar IR is mainly by:

(a) Nitrogen (b) Water vapour and CO_2

(c) Argon (d) Helium

Q20. Ultraviolet radiation is largely absorbed by atmospheric:

(a) Ozone (b) Oxygen only

(c) Dust (d) Methane

Q21. Scattering of sunlight by air molecules is:

(a) Mie scattering (b) Rayleigh scattering

(c) Reflection (d) Refraction

Q22. A pyranometer measures:

(a) Beam only (b) Global (beam + diffuse) radiation

(c) Sunshine hours (d) Wind speed

Q23. A pyrheliometer measures:

(a) Diffuse radiation (b) Beam (direct) radiation at normal incidence

(c) Global (d) Temperature

Q24. To measure DIFFUSE radiation, a pyranometer is fitted with a:

(a) Second dome (b) Shade ring/disc to block the beam

(c) Lens (d) Tracker

Q25. A pyrheliometer must:

(a) Lie flat (b) Track the sun (point at normal incidence)

(c) Be shaded (d) Face north

Q26. Hours of bright sunshine are recorded by a:

(a) Pyranometer (b) Campbell–Stokes sunshine recorder

(c) Anemometer (d) Barometer

Q27. Irradiance is expressed in:

(a) W/m² (b) MJ/day

(c) °C (d) m/s

Q28. Daily insolation (energy) is expressed in:

(a) W/m² (b) MJ/m²/day or kWh/m²/day

(c) N/m² (d) kg

Q29. Typical daily global insolation over most of India is about:

(a) 1–2 (b) 5–7

(c) 15–20 (d) 30 kWh/m²/day

Q30. India's total installed renewable-energy capacity (June 2026) is about:

(a) 100 GW (b) 175 GW

(c) 289 GW (d) 500 GW

Q31. Solar power is India's largest renewable source at about:

(a) 60 GW (b) 100 GW

(c) 162 GW (d) 250 GW

Q32. The scheme promoting solar pumps and solarisation for farmers is:

(a) PM-KUSUM (b) FAME

(c) UJALA (d) SATAT

Q33. PM-KUSUM Component-B provides:

(a) Grid solar plants (b) Standalone solar agricultural pumps

(c) Rooftop solar (d) EV subsidy

Q34. PV modules respond to only part of the spectrum because:

(a) Cells absorb all wavelengths (b) A cell's spectral response is limited (e.g. Si ~0.4–1.1 µm)

(c) There is no IR (d) UV is 90%

Q35. Reference solar instruments are traceable to the:

(a) WMO World Radiometric Reference (b) NIST clock

(c) ISO 9001 (d) OECD code

Answers

Q1 — (c) Petroleum/diesel

Q2 — (d) Solar

Q3 — (b) Indirect solar energy

Q4 — (b) Nitrogen fertiliser

Q5 — (b) Just outside the atmosphere

Q6 — (b) 1367 W/m² — *WMO value 1367 W/m² (modern satellite value ~1361 W/m²).*

Q7 — (b) 1.94 — *1.94 cal/cm²·min is the classic figure — it corresponds to 1353 W/m²; 1367 W/m² converts to ≈ 1.96.*

Q8 — (b) ±3.3% — *±3.3% over the year — Earth's elliptical orbit, nearest in early January.*

Q9 — (c) 5800 K

Q10 — (b) 0.48 µm — *5800 K → Wien peak ~0.48 µm.*

Q11 — (c) 47% — *Visible ~47% (PAR 0.4–0.7 µm).*

Q12 — (a) 7% — *UV ~7%, visible ~47%, IR ~46%.*

Q13 — (b) 0.4–0.7 µm

Q14 — (b) Beam (direct)

Q15 — (c) Beam + diffuse

Q16 — (b) 1/cos θ_z — *AM = 1/cos θz.*
Q17 — (b) Extraterrestrial (outside atmosphere)
Q18 — (c) AM1.5 — *AM0 space, AM1 overhead at sea level, AM1.5 the PV rating standard (~1000 W/m²).*
Q19 — (b) Water vapour and CO_2
Q20 — (a) Ozone
Q21 — (b) Rayleigh scattering
Q22 — (b) Global (beam + diffuse) radiation — *Pyranometer = global radiation.*
Q23 — (b) Beam (direct) radiation at normal incidence — *Pyrheliometer = beam (sun-tracking).*
Q24 — (b) Shade ring/disc to block the beam — *Pyranometer + shade ring = diffuse.*
Q25 — (b) Track the sun (point at normal incidence)
Q26 — (b) Campbell–Stokes sunshine recorder — *Campbell–Stokes = sunshine hours.*
Q27 — (a) W/m²
Q28 — (b) MJ/m²/day or kWh/m²/day
Q29 — (b) 5–7
Q30 — (c) 289 GW — *MNRE data to 30 June 2026: RE 288.58 GW.*
Q31 — (c) 162 GW — *Solar 162.15 GW (30 June 2026).*
Q32 — (a) PM-KUSUM
Q33 — (b) Standalone solar agricultural pumps — *PM-KUSUM B = standalone solar pumps.*
Q34 — (b) A cell's spectral response is limited (e.g. Si ~0.4–1.1 μm)
Q35 — (a) WMO World Radiometric Reference

Solar thermal collection, storage, PV conversion and SPV systems (Q36–Q70)

Q36. A flat-plate collector primarily collects:

(a) Beam only
(b) Global (beam + diffuse) radiation
(c) Diffuse only
(d) UV only

Q37. The glazing cover of a flat-plate collector works by:

(a) Reflecting all light
(b) The greenhouse effect — trapping long-wave re-radiation, cutting convection
(c) Cooling the plate
(d) Concentrating light

Q38. A flat-plate collector typically reaches temperatures up to about:

(a) 30 °C
(b) 100 °C
(c) 400 °C
(d) 1000 °C

Q39. Useful heat from a collector is Q =

(a) mc_pΔT (ṁ·c_p·(T_out-T_in))
(b) I·A only
(c) mgh
(d) ½mv²

Q40. Collector efficiency as plate temperature rises above ambient:

(a) Rises
(b) Falls (losses ∝ ΔT)
(c) Constant
(d) Zero

Q41. A selective absorber surface has:

(a) Low absorptance
(b) High absorptance and low emittance
(c) High emittance
(d) Mirror finish

Q42. Concentrating collectors use which radiation and require:

(a) Diffuse, no tracking
(b) Beam radiation and sun tracking
(c) Global, fixed
(d) UV, tracking

Q43. Concentration ratio (CR) is:

(a) Receiver ÷ aperture area
(b) Aperture ÷ receiver area
(c) Beam ÷ diffuse
(d) Temperature ratio

Q44. A parabolic trough is a:

(a) Point-focus collector
(b) Line-focus collector (~200–400 °C)
(c) Flat plate
(d) Evacuated tube

Q45. The highest temperatures (>500–1000 °C) are reached by:

(a) Flat plate
(b) Evacuated tube
(c) Point-focus dish / central receiver
(d) Trough

Q46. Sensible-heat storage stores energy by:

(a) Melting a material
(b) Raising a medium's temperature ($Q = mc_p\Delta T$)
(c) A chemical reaction
(d) Compression

Q47. The cheap, common sensible-heat storage medium for solar water heating is:

(a) Rock
(b) Water (high c_p)
(c) Sand
(d) Oil

Q48. A phase-change material (PCM) stores:

(a) Sensible heat only
(b) Latent heat at nearly constant temperature
(c) Electricity
(d) Momentum

Q49. For solar AIR heating/drying, a suitable storage is a:

(a) Battery
(b) Rock/pebble bed
(c) Capacitor
(d) Flywheel

Q50. PV converts sunlight into:

(a) Heat then electricity
(b) DC electricity directly (photovoltaic effect)
(c) AC directly
(d) Mechanical energy

Q51. The photovoltaic effect occurs at a semiconductor:

(a) Resistor
(b) p–n junction
(c) Capacitor
(d) Transformer

Q52. The band gap of silicon is about:

(a) 0.1 eV
(b) 1.1 eV
(c) 5 eV
(d) 12 eV

Q53. A single silicon cell produces about:

(a) 0.05 V
(b) 0.5–0.6 V
(c) 5 V
(d) 12 V

Q54. Connecting cells in SERIES increases the:

(a) Current
(b) Voltage
(c) Area
(d) Efficiency

Q55. Connecting modules in PARALLEL increases the:

(a) Voltage
(b) Current
(c) Band gap
(d) FF

Q56. Open-circuit voltage V_{oc} of a PV cell as temperature rises:

(a) Increases (b) Decreases

(c) Constant (d) Doubles

Q57. Short-circuit current I_{sc} is mainly proportional to:

(a) Temperature (b) Irradiance and cell area

(c) Voltage (d) Band gap

Q58. Fill factor FF equals:

(a) $V_{oc} \cdot I_{sc}$ (b) $(V_{mp} \cdot I_{mp})/(V_{oc} \cdot I_{sc})$

(c) I_{sc}/V_{oc} (d) $V_{mp}+I_{mp}$

Q59. The point of maximum V·I on the I–V curve is the:

(a) V_{oc} (b) I_{sc}

(c) Maximum power point (MPP) (d) Origin

Q60. An MPPT controller:

(a) Cools the panel (b) Keeps the array operating at its maximum power point

(c) Cleans dust (d) Tracks the sun mechanically

Q61. PV modules are rated at Standard Test Conditions:

(a) 500 W/m², 40 °C (b) 1000 W/m², 25 °C, AM1.5

(c) 1367 W/m², 0 °C (d) 800 W/m², 20 °C

Q62. The highest-efficiency common crystalline cell type is:

(a) Monocrystalline silicon (b) Polycrystalline

(c) Amorphous (d) CdTe

Q63. Thin-film PV (a-Si, CdTe, CIGS) is characterised by:

(a) Highest efficiency (b) Lower efficiency but cheap and flexible

(c) No degradation (d) Needing tracking

Q64. In 2026, commercial TOPCon modules reach efficiencies of about:

(a) 10–12% (b) 15–17%

(c) 22–23% (d) 40%

Q65. PERC modules in 2026 deliver about:

(a) 5% (b) 21–22%

(c) 30% (d) 45%

Q66. PV output typically falls with rising cell temperature by about:

(a) 0.01%/°C (b) 0.3–0.5%/°C

(c) 5%/°C (d) 20%/°C

Q67. The cheapest storage for a solar irrigation pump is usually to:

(a) Use large batteries (b) Pump water into a tank (store water, not charge)

(c) Use PCM (d) Use a flywheel

Q68. A solar pump system typically comprises:

(a) Array → MPPT/VFD → motor–pump (b) Array → boiler

(c) Array → engine (d) Battery → diesel

Q69. An inverter in an SPV system:

(a) Stores energy (b) Converts DC to AC for AC loads

(c) Tracks the sun (d) Cools cells

Q70. A major practical loss for field PV in India is:

(a) Too much diffuse light
(b) Dust deposition and shading (needs cleaning)
(c) Excess voltage
(d) Low band gap

Answers

Q36 — (b) Global (beam + diffuse) radiation

Q37 — (b) The greenhouse effect — trapping long-wave re-radiation, cutting convection — *Glazing traps long-wave re-radiation (greenhouse effect).*

Q38 — (b) 100 °C

Q39 — (a) $mc_p\Delta T$ ($\dot{m}\cdot c_p\cdot(T_{out}-T_{in})$)

Q40 — (b) Falls (losses ∝ ΔT) — *Efficiency falls as $(T_i - T_a)/I$ rises — the Hottel–Whillier–Bliss line.*

Q41 — (b) High absorptance and low emittance

Q42 — (b) Beam radiation and sun tracking — *Concentrators use beam only and track.*

Q43 — (b) Aperture ÷ receiver area — *CR = aperture/receiver.*

Q44 — (b) Line-focus collector (~200–400 °C) — *Line-focus (trough/Fresnel) 200–400 °C.*

Q45 — (c) Point-focus dish / central receiver — *Point-focus (dish/tower) > 500 °C.*

Q46 — (b) Raising a medium's temperature ($Q = mc_p\Delta T$)

Q47 — (b) Water (high c_p)

Q48 — (b) Latent heat at nearly constant temperature

Q49 — (b) Rock/pebble bed

Q50 — (b) DC electricity directly (photovoltaic effect)

Q51 — (b) p–n junction

Q52 — (b) 1.1 eV — *Si band gap 1.1 eV.*

Q53 — (b) 0.5–0.6 V — *Cell 0.5–0.6 V.*

Q54 — (b) Voltage

Q55 — (b) Current

Q56 — (b) Decreases — *V_{oc} falls with temperature.*

Q57 — (b) Irradiance and cell area — *I_{sc} ∝ irradiance.*

Q58 — (b) $(V_{mp}\cdot I_{mp})/(V_{oc}\cdot I_{sc})$

Q59 — (c) Maximum power point (MPP)

Q60 — (b) Keeps the array operating at its maximum power point

Q61 — (b) 1000 W/m², 25 °C, AM1.5 — *STC = 1000 W/m², 25 °C, AM1.5.*

Q62 — (a) Monocrystalline silicon

Q63 — (b) Lower efficiency but cheap and flexible

Q64 — (c) 22–23% — *As given in the source notes (July 2026) — TOPCon modules 22–23% (cells ~26%).*

Q65 — (b) 21–22% — *PERC 21–22%.*

Q66 — (b) 0.3–0.5%/°C — *Temperature coefficient ~0.3–0.5%/°C.*

Q67 — (b) Pump water into a tank (store water, not charge) — *Battery-less solar pumps store WATER in a tank (cheapest) — the PM-KUSUM standard.*

Q68 — (a) Array → MPPT/VFD → motor–pump — *Chain: array → MPPT/VFD → motor–pump.*

Q69 — (b) Converts DC to AC for AC loads

Q70 — (b) Dust deposition and shading (needs cleaning)

Wind mills; thermo-chemical conversion and densification of biomass

(Q71–Q105)

Q71. The power available in wind of speed V through area A is:

(a) ½ρAV² (b) ½ρAV³

(c) ρAV (d) ρA²V

Q72. Wind power is proportional to wind speed raised to the power:

(a) 1 (b) 2

(c) 3 (d) 4

Q73. Doubling the wind speed increases available power by:

(a) 2× (b) 4×

(c) 8× (d) 16×

Q74. The Betz limit (maximum power coefficient) is:

(a) 0.35 (b) 0.50

(c) 0.593 (16/27) (d) 1.0

Q75. Wind power is also proportional to:

(a) Rotor diameter (b) Rotor diameter squared (swept area)

(c) Blade number (d) Tower height cubed

Q76. Tip-speed ratio is the ratio of:

(a) Wind speed to rotor rpm (b) Blade-tip speed to wind speed

(c) Power to torque (d) Diameter to length

Q77. The classic multi-blade 'American' farm windmill is mainly used for:

(a) Grid power (b) Water pumping (high starting torque)

(c) Battery charging only (d) Lighting

Q78. A HAWT (horizontal-axis turbine) requires which mechanism to face the wind?

(a) Yaw (b) Pitch only

(c) None (d) Brake

Q79. The Darrieus and Savonius rotors are:

(a) HAWTs (b) VAWTs (vertical axis)

(c) Water turbines (d) Steam turbines

Q80. A Savonius rotor is a:

(a) Lift-type high-speed rotor (b) Drag-type, high-torque, self-starting rotor

(c) HAWT (d) Gas turbine

Q81. An advantage of VAWTs is that they:

(a) Are most efficient (b) Accept wind from any direction (no yaw)

(c) Need no support (d) Have no generator

Q82. Typical cut-in wind speed of a turbine is about:

(a) 0.5 m/s (b) 3–4 m/s

(c) 12 m/s (d) 25 m/s

Q83. The cut-out speed at which a turbine shuts down for safety is about:

(a) 5 m/s (b) 12 m/s

(c) 25 m/s (d) 50 m/s

Q84. Real wind turbines achieve a power coefficient of about:

(a) 0.05–0.10 (b) 0.35–0.45

(c) 0.60–0.70 (d) 0.90

Q85. India's installed wind capacity (2026) is about:

(a) 20 GW (b) 40 GW

(c) 56 GW (d) 100 GW

Q86. The distinguishing feature among thermo-chemical routes is the amount of:

(a) Water (b) Oxygen/air supplied

(c) Pressure (d) Catalyst

Q87. Direct combustion of biomass uses:

(a) No air (b) Partial air

(c) Excess (full) air (d) Pure nitrogen

Q88. Gasification is carried out with:

(a) Excess air (b) Partial (sub-stoichiometric) air

(c) No air (d) Pure oxygen only

Q89. The main combustible components of producer gas are:

(a) CO, H_2 and CH_4 (b) CO_2 and N_2

(c) O_2 and H_2O (d) SO_2 and NO

Q90. The calorific value of producer gas is roughly:

(a) 0.5 MJ/m³ (b) 4–6 MJ/m³

(c) 38 MJ/m³ (d) 120 MJ/m³

Q91. The gasifier that gives the cleanest gas suitable for engines is the:

(a) Updraught (b) Downdraught

(c) Open fire (d) None

Q92. Pyrolysis is thermal decomposition in the:

(a) Presence of excess air (b) Absence of oxygen

(c) Presence of water only (d) Presence of catalyst only

Q93. The three products of pyrolysis are:

(a) Char, bio-oil and gas (b) Only char

(c) Only gas (d) Ash and water

Q94. Fast pyrolysis is optimised to maximise:

(a) Char (b) Bio-oil (pyrolytic oil)

(c) Ash (d) CO_2

Q95. Slow pyrolysis / carbonization maximises:

(a) Bio-oil (b) Charcoal (char)

(c) Gas (d) Steam

Q96. Charcoal is produced by:

(a) Combustion (b) Carbonization (slow pyrolysis)

(c) Gasification (d) Fermentation

Q97. Biochar, a pyrolysis product, is also used as a:

(a) Fuel only (b) Soil amendment / carbon sink

(c) Fertiliser N source (d) Pesticide

Q98. Loose agro-residues have a bulk density of roughly:

(a) 40–150 kg/m³ (b) 600–1200 kg/m³

(c) 2000 kg/m³ (d) 5000 kg/m³

Q99. After densification, briquettes/pellets reach a density of about:

(a) 40–150 (b) 300

(c) 600–1200 (d) 3000 kg/m³

Q100. In binderless briquetting, the natural binder that softens under heat/pressure is:

(a) Cellulose (b) Lignin

(c) Starch (d) Silica

Q101. Typical pellet diameter is about:

(a) 1 mm (b) 6–12 mm

(c) 60–90 mm (d) 150 mm

Q102. A key benefit of densification is:

(a) Higher moisture (b) Higher bulk/energy density → cheaper transport & storage

(c) Lower CV (d) Larger size

Q103. Feedstock moisture before briquetting is typically reduced to about:

(a) 8–12% (b) 30%

(c) 50% (d) 70%

Q104. Biomass pellets help reduce which problem in North India?

(a) Soil salinity (b) Paddy-stubble burning (via co-firing)

(c) Groundwater (d) Wind erosion

Q105. Biomass is considered broadly carbon-neutral because:

(a) It has no carbon (b) The CO_2 released was recently fixed from the atmosphere

(c) It absorbs SO_2 (d) It emits O_2

Answers

Q71 — (b) ½ρAV³ — *$P = ½\rho AV^3$.*

Q72 — (c) 3

Q73 — (c) 8× — *$P \propto V^3$: $2^3 = 8$.*

Q74 — (c) 0.593 (16/27) — *$Cp \le 16/27 = 0.593$ (Betz).*

Q75 — (b) Rotor diameter squared (swept area) — *$P \propto A \propto D^2$.*

Q76 — (b) Blade-tip speed to wind speed

Q77 — (b) Water pumping (high starting torque)

Q78 — (a) Yaw

Q79 — (b) VAWTs (vertical axis)

Q80 — (b) Drag-type, high-torque, self-starting rotor

Q81 — (b) Accept wind from any direction (no yaw)

Q82 — (b) 3–4 m/s

Q83 — (c) 25 m/s

Q84 — (b) 0.35–0.45

Q85 — (c) 56 GW — *India wind ≈ 56 GW (31 Mar 2026), 4th largest globally.*

Q86 — (b) Oxygen/air supplied — *Routes ranked by oxygen — combustion (excess) → gasification (partial) → pyrolysis (none).*

Q87 — (c) Excess (full) air

Q88 — (b) Partial (sub-stoichiometric) air

Q89 — (a) CO, H_2 and CH_4 — *Producer gas = $CO + H_2 + CH_4$.*

Q90 — (b) 4–6 MJ/m³ — *CV 4–6 MJ/m³.*

Q91 — (b) Downdraught — *The downdraught gasifier gives engine-grade gas.*

Q92 — (b) Absence of oxygen
Q93 — (a) Char, bio-oil and gas — *Pyrolysis gives char + bio-oil + gas.*
Q94 — (b) Bio-oil (pyrolytic oil) — *Fast pyrolysis → bio-oil.*
Q95 — (b) Charcoal (char) — *Slow pyrolysis/carbonization → charcoal.*
Q96 — (b) Carbonization (slow pyrolysis)
Q97 — (b) Soil amendment / carbon sink
Q98 — (a) 40–150 kg/m³ — *Densification raises 40–150 → 600–1200 kg/m³.*
Q99 — (c) 600–1200
Q100 — (b) Lignin — *Lignin binds.*
Q101 — (b) 6–12 mm — *Pellets 6–12 mm; briquettes 60–90 mm.*
Q102 — (b) Higher bulk/energy density → cheaper transport & storage
Q103 — (a) 8–12% — *Dry to ~8–12%.*
Q104 — (b) Paddy-stubble burning (via co-firing)
Q105 — (b) The CO_2 released was recently fixed from the atmosphere

Bioconversion, biogas, alternate fuels and energy analysis (Q106–Q140)

Q106. Biogas is produced by:
(a) Aerobic composting
(b) Anaerobic digestion of organic matter
(c) Combustion
(d) Gasification

Q107. The four stages of anaerobic digestion end with:
(a) Hydrolysis
(b) Acidogenesis
(c) Acetogenesis
(d) Methanogenesis

Q108. Methane content of raw biogas is about:
(a) 10–20%
(b) 30–40%
(c) 55–65%
(d) 90–95%

Q109. The major non-combustible component of raw biogas is:
(a) CO_2 (35–44%)
(b) N_2
(c) O_2
(d) H_2

Q110. Optimum temperature for mesophilic digestion is about:
(a) 10 °C
(b) 35 °C
(c) 60 °C
(d) 90 °C

Q111. The ideal C:N ratio for biogas feedstock is about:
(a) 5:1
(b) 20–30:1
(c) 60:1
(d) 100:1

Q112. The KVIC biogas plant uses a:
(a) Fixed masonry dome
(b) Floating steel/FRP gas holder
(c) Balloon
(d) Rock bed

Q113. An advantage of the floating-drum plant is:
(a) No moving parts
(b) Constant gas pressure
(c) Lowest cost
(d) No corrosion

Q114. The Deenbandhu plant is a:
(a) Floating drum
(b) Fixed-dome plant (no moving part, cheaper)
(c) Portable bag
(d) Community digester only

Q115. A drawback of the fixed-dome plant is:

(a) Corrosion of drum
(b) Variable gas pressure
(c) Moving parts
(d) High steel cost

Q116. The spent slurry from a biogas plant is valuable as:

(a) Fuel
(b) Nitrogen-rich manure
(c) Pesticide
(d) Cement

Q117. Ethanol for blending is produced by:

(a) Transesterification
(b) Fermentation of sugars/starch by yeast
(c) Pyrolysis
(d) Gasification

Q118. Biodiesel is produced by:

(a) Fermentation
(b) Transesterification of oils/fats with an alcohol + catalyst
(c) Distillation of crude
(d) Anaerobic digestion

Q119. Transesterification of vegetable oil with methanol gives biodiesel plus:

(a) Water
(b) Glycerol
(c) Ethanol
(d) CO_2

Q120. The main reason for converting vegetable oil to biodiesel is to:

(a) Increase viscosity
(b) Reduce viscosity so it runs in diesel engines
(c) Add colour
(d) Raise density

Q121. FAME (biodiesel) using ethanol instead of methanol yields:

(a) Methyl esters
(b) Ethyl esters
(c) Glycerol only
(d) Producer gas

Q122. Common alkali catalyst for transesterification is:

(a) H_2SO_4 only
(b) KOH/NaOH
(c) NaCl
(d) $CaCO_3$

Q123. Compressed Bio-Gas (CBG) per IS 16087:2016 must contain minimum methane of:

(a) 50%
(b) 75%
(c) 90%
(d) 99%

Q124. CBG is compressed to a pressure of about:

(a) 5 bar
(b) 50 bar
(c) 250 bar
(d) 1000 bar

Q125. The Indian scheme promoting CBG plants is:

(a) SATAT
(b) FAME
(c) UJALA
(d) PM-KUSUM

Q126. Ethanol as an SI-engine fuel has, relative to petrol:

(a) Higher octane, lower calorific value
(b) Lower octane
(c) Higher CV
(d) No difference

Q127. Biodiesel relative to diesel typically has:

(a) Lower cetane
(b) Higher cetane and good lubricity
(c) No cetane
(d) Higher sulphur

Q128. A challenge common to alternative fuels is:

(a) Material compatibility and calorific-value adjustment
(b) Too much power
(c) No storage need
(d) Zero cost

Q129. DIRECT energy in agriculture includes:

(a) Fertiliser manufacture
(b) Diesel, electricity, human and animal power
(c) Seed embodied energy
(d) Machinery manufacture

Q130. The LARGEST indirect (embodied) energy input in most crops is:

(a) Seed
(b) Nitrogen fertiliser
(c) Pesticide
(d) Machinery

Q131. Human, animal, biomass and solar energy are classified as:

(a) Non-renewable
(b) Renewable
(c) Commercial only
(d) Fossil

Q132. Diesel, electricity and fertiliser are examples of:

(a) Non-commercial energy
(b) Commercial energy
(c) Renewable energy
(d) Muscle power

Q133. Energy ratio (energy-use efficiency) is:

(a) Input ÷ output
(b) Output ÷ input
(c) Output − input
(d) Output × input

Q134. An energy ratio greater than 1 means:

(a) Net energy loss
(b) Net energy gain
(c) No output
(d) Zero input

Q135. Specific energy is defined as:

(a) Input ÷ yield (MJ/kg)
(b) Yield ÷ input
(c) Output ÷ area
(d) Input × yield

Q136. Inputs 18,000 MJ/ha and grain energy output 58,800 MJ/ha give an energy ratio of about:

(a) 1.3
(b) 2.0
(c) 3.3
(d) 5.0

Q137. For the same crop (4 t/ha, 18,000 MJ/ha), specific energy is about:

(a) 1.5
(b) 4.5 MJ/kg
(c) 14.7
(d) 45

Q138. Energy productivity is the reciprocal of specific energy, in units of:

(a) MJ/kg
(b) kg/MJ
(c) MJ/ha
(d) kg/ha

Q139. An energy AUDIT is:

(a) A financial statement
(b) A systematic inventory of all energy inflows and outflows converted to MJ
(c) A soil test
(d) A yield forecast

Q140. A way to improve the energy ratio of a cropping system is to:

(a) Increase nitrogen use
(b) Replace diesel pumping with solar and reduce tillage
(c) Add more machinery
(d) Irrigate more often

Answers

Q106 — (b) Anaerobic digestion of organic matter

Q107 — (d) Methanogenesis — *Digestion ends in methanogenesis.*

Q108 — (c) 55–65% — *Biogas 55–65% CH_4.*

Q109 — (a) CO_2 (35–44%)

Q110 — (b) 35 °C — *Optimum ~35 °C.*

Q111 — (b) 20–30:1 — *C:N 20–30.*
Q112 — (b) Floating steel/FRP gas holder — *KVIC floating drum vs Deenbandhu fixed dome — the classic comparison.*
Q113 — (b) Constant gas pressure — *Floating drum gives constant pressure; the steel drum corrodes.*
Q114 — (b) Fixed-dome plant (no moving part, cheaper) — *Deenbandhu: no moving part, cheap.*
Q115 — (b) Variable gas pressure — *Fixed dome: variable pressure.*
Q116 — (b) Nitrogen-rich manure
Q117 — (b) Fermentation of sugars/starch by yeast
Q118 — (b) Transesterification of oils/fats with an alcohol + catalyst — *Biodiesel by transesterification (triglyceride + methanol + KOH → methyl esters + glycerol).*
Q119 — (b) Glycerol
Q120 — (b) Reduce viscosity so it runs in diesel engines — *Transesterification reduces viscosity.*
Q121 — (b) Ethyl esters — *Ethanol → ethyl esters.*
Q122 — (b) KOH/NaOH
Q123 — (c) 90% — *CBG IS 16087:2016 ≥ 90% methane.*
Q124 — (c) 250 bar — *Compressed to ~250 bar.*
Q125 — (a) SATAT — *SATAT scheme (as given in the source notes: 132 plants / 920 TPD commissioned, July 2026).*
Q126 — (a) Higher octane, lower calorific value
Q127 — (b) Higher cetane and good lubricity
Q128 — (a) Material compatibility and calorific-value adjustment
Q129 — (b) Diesel, electricity, human and animal power
Q130 — (b) Nitrogen fertiliser
Q131 — (b) Renewable
Q132 — (b) Commercial energy
Q133 — (b) Output ÷ input
Q134 — (b) Net energy gain
Q135 — (a) Input ÷ yield (MJ/kg)
Q136 — (c) 3.3 — *Energy ratio = 58,800/18,000 = 3.3.*
Q137 — (b) 4.5 MJ/kg — *Specific energy = 18,000/4000 = 4.5 MJ/kg.*
Q138 — (b) kg/MJ — *Productivity = 1/4.5 = 0.22 kg/MJ.*
Q139 — (b) A systematic inventory of all energy inflows and outflows converted to MJ
Q140 — (b) Replace diesel pumping with solar and reduce tillage